EXPERIMENT 12

Prompt Evaluation, Comparison and Refinement using LLM Responses

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AIM

To demonstrate the comparison of multiple prompts using evaluation criteria and refinement of the selected prompt.

PROCEDURE

1. Select a text-generation task: explain Machine Learning.
2. Create three prompts with different levels of detail.
3. Display each prompt.
4. List the evaluation criteria: relevance, accuracy, completeness, clarity, and format adherence.
5. Prepare a refined prompt with clear instructions.
6. Display the refined prompt and a sample final response.

PROGRAM

```
print("Prompt Evaluation, Comparison and Refinement")

prompts = [
    "Explain Machine Learning.",
    "Explain Machine Learning in simple words.",
    "Explain Machine Learning with examples."
]

for index, prompt in enumerate(prompts, start=1):
    print("\nPrompt", index, ":", prompt)

print("\nEvaluation Criteria:")
print("1. Relevance")
print("2. Accuracy")
print("3. Completeness")
print("4. Clarity")
print("5. Format Adherence")

refined_prompt = (
    "Explain Machine Learning in simple words "
    "with one real-world example."
)

print("\nRefined Best Prompt:")
print(refined_prompt)

print("\nSample Final Output:")
print("Machine Learning is a branch of AI that enables")
print("computers to learn from data and make predictions.")
print("Example: Email spam detection.")
```

OUTPUT

```
print("\nFinal Output:")
print("Machine Learning is a branch of AI that enables")
print("computers to learn from data and make predictions.")
```

```
Prompt Evaluation and Refinement

Prompt 1:
Explain Machine Learning.

Prompt 2:
Explain Machine Learning in simple words.

Prompt 3:
Explain Machine Learning with examples.

Evaluation Criteria:
1. Relevance
2. Accuracy
3. Completeness
4. Clarity
5. Format Adherence

Refined Prompt:
Explain Machine Learning in simple words with examples.

Final Output:
Machine Learning is a branch of AI that enables
computers to learn from data and make predictions.
```

RESULT

Thus, the prompts were compared using the specified criteria and a refined prompt was demonstrated.